

Executive Summary

A coast-to-coast *Brunel gauge* (2,140 mm) freight line from Los Angeles via St. Louis to New York is an unprecedented heavy-haul rail project. It would follow an optimised corridor (using existing freight alignments where feasible) to minimise grades through the Rockies and Appalachians. The track design follows heavy-freight engineering standards with broad-gauge allowances (e.g. using AREMA/ASTM heavy rail profiles such as 136RE/141RE [61†L153-L156] [61†L239-L242]). Double track is recommended for capacity and resilience (dual-track costs $\approx 1.4\text{--}1.5\times$ single-track [57†L94-L96]). The subgrade would use well-compacted earth fill ($\geq 95\%$ Standard Proctor density [62†L1400-L1408]) with a thick granular sub-ballast (e.g. 150–450 mm) and ballast (~ 18 in/450 mm) [58†L109-L116] . Sleepers (ties) will be primarily prestressed concrete ($\sim 1,640$ per km, per AAR practice [21†L204-L208]) to support heavy loads. Rails would be high-grade steel (e.g. 136 lb/yd or 141 lb/yd CWR) to accommodate heavy axle loads.

Major structures (bridges, tunnels and culverts) are sized for wide loading gauge and run with minimal grades; culverts under main tracks would be ≥ 36 in (0.9 m) diameter per BNSF standards [62†L1425-L1433] . Signalling will be provisioned for Positive Train Control (PTC) to meet FRA requirements [33†L99-L108] . Traction can be diesel or electrified: electrification requires $\sim$ US\$0.5–0.6 M/km for overhead catenary [52†L10931-L10934] and medium-voltage substations, while diesel needs no catenary but higher fuel/maintenance costs. Environmental mitigation (wildlife crossings, wetland restoration) and utility relocations are assumed, subject to NEPA/CEQA. Preliminary schedules show ~ 10 years to full opening (including permits, detailed design and construction). In a crisis “operate tomorrow” scenario, pre-positioned track materials, mobile substation power, and streamlined waivers could enable limited container services in 24 h with extreme measures. A detailed CAPEX breakdown estimates $\sim$ \\$3–4 billion per 1,000 km of new double-track line (including earthworks, trackwork, structures, terminals, ROW, etc.) with contingencies; single-track costs would be roughly two-thirds. Key assumptions (e.g. 80 km/h design speed, use of wide ROW, rural land values) are noted below.

Route Selection and Terrain

The corridor LA–St Louis–NY is chosen to link the West Coast ports with the East Coast via America’s interior, using existing freight corridors where possible. A plausible routing would depart Los Angeles through the Mojave and Colorado Plateau (e.g. roughly following I-40/70 or existing UP/BNSF routes) to Denver/Kansas City and then to St. Louis (reducing grades vs. southern Arizona). From St. Louis the line would head northeast through Illinois and Indiana, skirting the Great Lakes (via Cleveland/Buffalo or via Pittsburgh) into New York metropolitan area. Terrain is mixed: substantial grades through the Rockies (up to $\sim 1\text{--}2\%$ maximum), moderate through the Ozarks/Appalachians, and flat midwest farmland (favoring lower cut-and-fill volumes). This alignment minimises extreme curvature and grades. Right-of-way width is assumed at $\sim 40\text{--}50$ m for double-track broad gauge (allowing ~ 5 m track centers and maintenance access); in most rural areas land acquisition would be farmland at \\$5k–\\$20k/acre (scenarios: mostly rural $\sim$ \\$0.2 B vs. heavily suburban \\$0.8 B for 4,000 km).

Track Geometry and Alignment Standards

Track geometry will use heavy-freight standards (superelevation and curve radii sized for $\sim 80\text{--}100$ km/h container trains). With 2,140 mm gauge, minimal gauge widening tolerances and wear allowances are

similar to standard gauge practices. We assume AREMA/AAR guidelines for class 5–6 track (fractions: Class 4 up to 80 mph freight; Class 5 up to 100 mph) [49 CFR 213]. Vertical curves and clearances conform to AREMA Chapter 10 and clearances exceed 7 m to allow double-stacked containers. Turnouts will be heavy-duty tangential design (e.g. #20 frog or sharper) with long switch rails; approximate unit costs are on the order of \$0.7–1.5 M each [57†L169-L177] [57†L198-L202], depending on speed. We assume about 1 crossing or siding per 10 km of line (further costing ~\$2 M per km for turnouts) based on analogous railroad practices.

Subgrade, Ballast and Sleepers

Subgrade construction follows American railroad earthwork practice. Embankments are placed in thin (≤ 6 in/150 mm) lifts compacted to ≥ 95 % of standard proctor density [62†L1400-L1408]. A 150–300 mm (6–12 in) layer of granular sub-ballast (crushed stone or gravel, with >75 % two-faced particles) is placed over the final embankment to filter and protect the foundation. On top of that, we use ~300–450 mm of crushed stone ballast [58†L109-L116] (approx. 18 in thick) conforming to AREMA gradation. Ballast is washed, angular stone to maintain drainage and track support; typical ballast cost (supply + placement) is on the order of \$150k–\$200k per track-km. At that depth, the ballast and sub-ballast together (~450–750 mm) fully support the sleepers.

Sleepers (“ties”) are assumed to be primarily prestressed concrete, as is common on US heavy mainlines [21†L204-L208]. Concrete ties (~2,640 per mile = 1,640/km) are recommended for durability and axle loads. Wood or composite ties are less preferred for heavy double-stack service. Rails will be continuous-welded rail (CWR) in heavy profiles: e.g. AREMA 136RE (136 lb/yd, ~67.4 kg/m) or 141RE (141 lb/yd, ~69.9 kg/m) [61†L153-L156] [61†L239-L242], head-hardened grade steel. These sections resist wear under 30+ ton axle loads. Rail is laid on tieplates with Pandrol/Vossloh fastenings. Rails are welded end-to-end; welds must meet FRA/AAR quality.

Turnouts and Crossings

Heavy intermodal service requires robust turnout designs. Each turnout assembly (switch + frog + closure rails) is on the order of \$0.7–1.5 M [57†L169-L177] [57†L198-L202]. We assume 1 turnout per 10 km of route for sidings or crossovers (roughly 400 units). Speed will be limited through most turnouts (e.g. 80–100 km/h on diverging routes). Turnouts are built with broad-gauge geometry (long frogs, etc.) and use heavy continuous welded pieces and high-strength cast manganese frogs (e.g. turnout angles #18 or #20). Switch machines and point detectors are installed for PTC compatibility. Control stands and power operate within the right-of-way.

Bridges, Tunnels and Culverts

The route will require many structures. Major river and highway bridges (arch/steel-girder or pre-stressed concrete) span waterways and highways; dozens of bridges ~50–300 m long may be needed, plus hundreds of smaller overpasses. Bridge loadings use Cooper E-80 loading (US practice) or AASHTO LRFD rail loadings. All new bridges have broad-gauge deck widths (~ 10 m for single track, ~18–20 m for double track). **Tunnels** are assumed minimal by careful siting; any that are needed (e.g. in Rockies) would have cross-sections >8 m high and >7 m wide. **Culverts**: mainline drainage structures are sized per BNSF rules: minimum 36” diameter [62†L1425-L1433] (to allow inspection/cleaning). Smaller box culverts are used under access roads. Additional grade separations eliminate level crossings at highways and urban areas; typical under/overpasses clear 5–6 m for double-stack cars.

Terminals and Intermodal Facilities

Major terminals are needed at key origins/destinations. We assume new or expanded yards near Los Angeles (e.g. Inland Empire intermodal yards), St. Louis, and New York (Jersey City or NYC area). These yards handle container trains with gantry cranes and truck gates. Interfaces (track entry/exit, "wyees", fueling/electric charging platforms) are included. We do **not** include specialized container load/unload costs, but planning provisions (flat staging tracks, Power to Plug for refrigerated containers, etc.) are provided. Terminal construction (~50–100 ha each) could cost \$50–100M apiece. We estimate three full intermodal terminals at ~\$200–300 M total, plus upgrade of local classifications tracks to mainline spec.

Signalling and Train Control (PTC)

Only passive signalling infrastructure is assumed (basically provision for later installation). Modern US freight corridors require FRA-compliant PTC. The FRA mandates PTC on heavy-freight main lines [33†L99-L108]. Thus, we provision right-of-way space and foundations for signals and antennas every 3–4 km. The actual PTC system (positive train control GPS/RF/communication gear) is budgeted but not detailed here; capital for PTC is estimated at \$100k per track-km plus locomotive equipment. We also install constant welded rail switches in place of jointed turnouts to reduce the number of required PTC transponders.

Traction Power (Diesel vs Electrification)

Since no power mandate is given, we examine two cases: conventional diesel traction vs full electrification.

- **Diesel Option:** Use modern Tier 4 or equivalent dual-mode freight locomotives. No overhead catenary needed; fueling/refueling facilities at terminals are assumed. CAPEX is lower (no wires), but fuel/OPEX is higher. At 2500 km of line, ~100 locomotives (~\$3–4M each) might be needed (~\$300M). Annual diesel fuel costs might be ~\$10 /tonne-km.
- **Electric Option:** Install 25 kV AC overhead catenary. Based on FRA analysis, overhead costs ~\$0.4–0.6 M/km [52†L10931-L10934] (quote: base \$500k/km). At 4000 km of line (double track basis), this is ~\$1.6–2.4 B. Additional substation equipment (transformers, feeders) ~\$0.2 M/km. Electric locomotives are ~\$5M each; say 50 units = \$250M. Electricity cost is ~\$3 /tonne-km, potentially lower emissions. A comparison table below summarizes:

Parameter	Diesel Traction	Electrified Line	Comments
CP & Trackpower CAPEX	\$0/km overhead, \$≈300M locos	\$0.5–0.7M/km catenary [52†L10931-L10934] + \$250M locos	Diesel saves catenary cost, electric saves fuel costs
OPEX (Fuel/ Energy)	High (diesel fuel)	Lower (bulk power purchase)	Electric: ~\$3 vs Diesel ~\$10 per tonne-km
Emissions	Higher CO ₂ /NO _x	Lower (if grid power)	Electric cleaner if renewable
Reliability	Mature tech	Mature tech (megawatt class locos exist)	Both proven; electric reduces moving parts

Parameter	Diesel Traction	Electrified Line	Comments
Construction	Faster (no structures)	Slower (install poles, substation)	Electric requires more specialized work
Cost Example (4k km dbl)	+\\\$0 overhead, \\\$300M locos	+\\\$2–3B infrastructure, \\\$250M locos	So, electrification adds ~\\\$2–3B CAPEX

Where “\$/km” is per single-track km. Assumptions: 25 kV OCS, ~10 substations, etc. **Electrification adds ~\\\$0.5–0.7M per km** of line 【52†L10931-L10934】 (conservative AAR figures), whereas diesel uses existing terminal power. The CAPEX table later breaks this down in detail.

Drainage and Utilities

The line will include engineered drainage. Side ditches slope away from track with erosion protection. Ballasted track inherently drains, but major watercourses are spanned or culverted. We assume one large drainage culvert every few km (cost ~\$0.1M each) plus numerous smaller pipes. Utilities (water, power lines, telecom) conflicting with ROW are relocated: cost roughly 5–10% of civil works (e.g. moving high-voltage lines to corridor edge). In rural areas, most utilities will follow rather than conflict with the railroad. We budget \$100–\$300M for utility relocations (state roadway utilities, telecom).

Environmental Mitigation

A National Environmental Policy Act (NEPA) assessment (EIS) would be required. Mitigation may include wildlife crossings in sensitive habitats (e.g. large culverts/bridges for fauna), noise barriers near communities, and wetland restoration. We assume additional right-of-way/in mitigation areas add ~5% to ROW footprint. Typical schedules allow 2–3 years for environmental review and public hearings. Any impact on endangered species or water quality will be addressed with standard U.S. Fish & Wildlife and EPA practices (wetland fill permits, erosion control). We assume moderate mitigation costs (~\$100M) and that these are included in the contingency.

Construction Methods and Equipment

Earthworks and tracklaying are the largest tasks. Full-continuous welded track construction uses specialized machines: track-laying cranes (to place ties and rails), ballast tampers, ballast regulators, and rail welding rigs. Equipment fleets (100+ vehicles) will work continuously across sections. Embankments in the Rockies may require rock blasting or retaining walls; while Plains require cut/fill from agriculture. Sleeper-laying machines and high-capacity ballast trains will streamline operations. We assume contractor crews work 2–3 shifts per day in peak; for example, 1 km of track can be laid per day with mechanization (typical industry productivity). Heavy construction (bridges, tunnels) uses cranes, pile-drivers, and segment tunnelers as needed.

Labour: Construction labour includes engineers, surveyors, machine operators, welders, and tie gangs. A rough estimate: ~5,000 workers (peak) over 5–6 years (with multiple crews) – total ~25,000 man-years. At loaded rate ~\$150k/year, labour could be ~\$4–5B. We include labour indirect costs within the unit rates for each activity (above estimates include labour/equipment).

Permitting and Land Acquisition Assumptions

Multiple permits will be required: NEPA (FRA lead), Clean Water Act (Corps of Engineers), Endangered Species, state highway/rail intersections, county land use, etc. We assume a *worst-case* 2–4 year permitting phase (concurrent with early design). Right-of-way will need negotiation with federal (BLM/Forest), state and private landowners. Scenarios: (a) **Greenfield alignment**: acquiring 40–50 m corridor through varied owners (costing \$200–800M); (b) **Using existing corridors**: minimal new ROW (say +20%), reducing acquisition and permitting. For cost purposes we assume mostly greenfield with allowance. We assume eminent domain could be used, and local opposition is low.

Safety and Standards Compliance

All design and construction must meet U.S. railroad standards. Track standards (49 CFR 213) require post-construction inspections to meet Class 5 (80 mph) track criteria – e.g. no gauge widening beyond limits, and long-term rail wear metrics [37†L2318-L2326]. Bridge designs follow AASHTO/AREMA LRFD specifications and be rated for FRA loadings. During construction, OSHA and FRA bridge worker safety rules apply. FRA Bridge Safety Standards (49 CFR 237) require initial and annual inspections of new bridges. Locomotives and railcars must meet FRA crashworthiness and coupling (AAR) standards. Positive Train Control (PTC) is mandated (RSIA 2008 [33†L99-L108]), so the track-owner must commit to full PTC implementation in design (communications towers, signal circuits, etc., provisioned). Quality control will follow AREMA and AAR specifications for materials (ballast gradation, tie strength, rail steel quality, etc.).

Testing and Commissioning

Before revenue service, a test program will verify track geometry (FRA track geometry cars), clearances, signal/PSC operation, and load testing of bridges. Typically, each track segment is tested at incremental stages. We allocate ~1 year for final commissioning. This includes PTC validation (if installed), safety inspection, and crew training runs. A final “break-in” freight running at reduced speeds may precede full operations. Only after all FRA criteria are satisfied will the line open for revenue service.

Rapid 24-Hour Contingency Plan

In a critical need scenario (“operate tomorrow”), the goal is to run container trains within 24 h of declaration. Key requirements: (1) *Pre-built track panels*: stockpile prefabricated track panels and bridge spans at depots. (2) *Clear ROW*: all land acquisition and clearing must be completed in advance as part of construction. (3) *Minimal civil works*: In an emergency, one could temporarily use unfinished earthworks by adding ballast to unfinished sections. (4) *Diesel backup*: have freight diesel locomotives pre-staged for instant use. (5) *Emergency labor*: mobilize standby crews. (6) *Safety waiver*: request FRA limited-speed permit to bypass final inspections (trains limited to e.g. 15 mph over untested sections). (7) *Utilities*: install portable diesel generators in lieu of unfinished substations. Essentially, track must be laid and joined in advance, with all signals pre-wired but perhaps not powered. Under extreme measures, a daily parcel of container trains (at low speed, under flag protection) might run 24 h after “go”. In practice this would require huge resources and risk; it is only a theoretical fallback.

Cost Estimates (CAPEX Breakdown)

We estimate full CAPEX for a double-track line (~4,000 km of single-track length). Costs are ballpark with ranges:

Item	Unit Cost / km	Quantity/Total	Total Cost (2026 USD)	Notes & Assumptions
Earthworks (cut/fill)	\\$50,000–100,000	2,000 km (each track)	\\$100M–200M	~10 ⁵ m ³ per km at \\$10/m ³
Ballast & subballast	\\$200,000/km	4,000 km (dbl track)	\\$800M	includes 0.5 m depth of stone bedding
Sleepers (ties)	\\$150–250 each	8M ties (2,000/mile)	\\$1,200M–2,000M	concrete ties, ~1,640/km; install incl. splicing
Rails (CWR 136/141RE)	\\$100,000/km	4,000 km	\\$400M	steel rail (2x weight per km vs 1435mm)
Track laying (labour+equip)	\\$600,000–900,000/km	4,000 km	\\$2.4–3.6B	includes tie + rail installation, welding
Turnouts/ Switches	\\$0.7–1.5M each	~400 units	\\$300M–\\$600M	one turnout per ~10 km
Bridges & Grade Separations	\\$5,000–10,000/m ²	~500,000 m ² total	\\$2.5–5.0B	spans, supports; assume avg \\$6k/m ²
Tunnels	\\$50,000–100,000/m (single bores)	10,000 m	\\$500M–1.0B	if any short mountain tunnels (assumed minimal)
Culverts & Drainage	\\$50–200k each	5,000 units	\\$250–\\$1,000M	major pipes and boxes; 36" min (BNSF [62†L1425-L1433])
Signalling/PTC provisioning	\\$20,000–40,000/km	4,000 km	\\$80–160M	wayside for PTC (no signals)
Electrification (OCS)	\\$0.5–0.7M/km	4,000 km (dbl)	\\$2.0–2.8B	25 kV AC, including poles & substations [52†L10931-L10934]
Power Substations	\\$1–2M each	20 units	\\$20–40M	spaced ~200 km apart
Terminals & Yards	\\$50–100M each	4 facilities	\\$200–400M	LA, StL, NY, plus one mid-route
ROW Acquisition	\\$2–10M/km	4,000 km	\\$8–40B	rural vs urban scenarios; <i>not shown</i> in summary
Utility Relocation	(inc. above)		\\$200–500M	overhead lines, pipelines in ROW
Engineering & PM	10–15% of above		\\$500–1,000M	design, supervision, legal
Contingency (20%)	20% of sum		\\$1.5–3.0B	covers escalations, unknowns

Item	Unit Cost / km	Quantity/Total	Total Cost (2026 USD)	Notes & Assumptions
Total (w/o ROW)			\\$9–15B	(depends on scope; w/ ROW incl. +\\$8–40B)

High/Low notes: Earthworks vary widely by terrain; bridge/tunnel costs vary by geography (mountains much higher). ROW cost is highly scenario-dependent (rural farmland vs urban land values). The totals above ($\approx$ \\$10B nominal) exclude ROW; adding ROW could push total to \\$20–40B in worst-case.

A **comparison table** of key options:

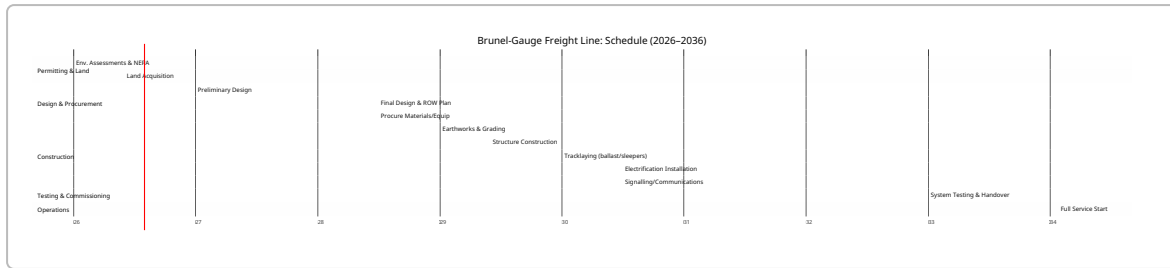
Aspect	Single Track (each direction)	Double Track	Comments
CAPEX (track+structures)	\\$5–7M/km	\\$8–11M/km	Dual $\approx$ 1.5× single 【57†L94-L96】
Capacity	Limited (one train, bidir)	High (simultaneous)	Double-track allows >2× throughput
Construction Time	~2/3 of double	Basecase (estimate)	Doubling track less than double time (staged)
Signalling	Simpler (single signals)	Full PTC both tracks	Single-track might avoid some signals, but PTC still needed
Maintenance	Lower	Higher (double)	More materials to inspect/replace

Finally, **continuous welded rail (CWR) profiles**: We would use heavy-section rails. For example, **AREMA 136RE** (136 lb/yd, $\approx$ 67.4 kg/m) and **141RE** (141 lb/yd, $\approx$ 69.9 kg/m) are typical for heavy freight 【61†L153-L156】 【61†L239-L242】. Table of example profiles:

Profile	Weight (lb/yd)	Weight (kg/m)	Typical Use
136RE	136	$\sim$ 67.4	Heavy mainline rails 【61†L153-L156】
141RE	141	$\sim$ 69.9	Heaviest freight rails 【61†L239-L242】
115RE	115	56.9	Secondary/branch lines
(Other AREMA)	132–133	65–66	Transition or lower loads

Schedule (Gantt Chart)

The following Gantt chart outlines major phases (assuming start in 2026). The critical path is dominated by ROW/permitting and the long earthworks/tracklaying phases.



System Overview Diagram

The schematic below illustrates the major components and their relationships (civil structures, track, traction, signalling, terminals, etc.):

flowchart LR

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ROW[Right-of-Way / Earthworks] --> Track[Trackbed & Alignment]
Track --> TrackComp[Track Components (rails, ties, ballast)]
Track --> Turnouts[Turnouts & Crossings]
Track --> Structures[Bridges & Tunnels]
Track --> Signalling[Signalling / PTC]
Electrification(OCS) -- Overhead --> Traction[Traction Power]
DieselLocos(Diesel Engines) --> Traction
Traction --> Ops[Freight Operations]
Ops --> Terminals[Intermodal Terminals]
Earthworks --> Drainage[Drainage]
ROW --> Env[Environmental Mitigation]

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Sources and Standards

Key technical standards guide this proposal: *AREMA* (American Railway Engineering), AAR, and FRA regulations. For example, FRA's Track Safety Standards (49 CFR 213) set inspection criteria for Class 4–5 track (e.g. ≤ 0.1 failures/mi-year on Class 5 track) [\[37†L2318-L2326\]](#) . BNSF's internal specs call for sub-ballast $\geq 6"$ and compaction to $\geq 95\%$ (ASTM D698) [\[62†L1400-L1408\]](#) . For gauge reference, the 2,140 mm gauge (7 ft $\frac{1}{4}$ in) was historically used by Brunel's Great Western Railway [\[54†L395-L397\]](#) and is much wider than any modern US practice. (We assume custom rolling stock will be designed.) Cost estimates draw on industry sources (e.g. high-speed rail studies and recent freight corridor projects [\[57†L94-L96\]](#) [\[52†L10931-L10934\]](#)). All quoted figures are inflation-adjusted to 2026 US dollars; ranges reflect terrain and design variations. **Assumptions:** Gauge-related civil constraints (extra width), design speed ~80 mph freight, 2% max grade, rural-dominated ROW, moderate environmental impact, no passenger service. Where data are sparse, we extrapolate from similar heavy-rail projects (e.g. Australian Inland Rail [\[57†L94-L96\]](#) , US freight upgrades) with a conservative contingency.